From Concern Detection to Promotion Control: A Protocol for Fail-Closed Governance of Autonomous Research Artifacts

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Abstract

Autonomous research systems increasingly produce complete-looking manuscripts and run bundles, yet review components can identify serious scientific defects without preventing a paper-ready decision. We formulate this gap as *promotion governance*: deciding whether an artifact bundle should be promoted, reviewed, downgraded, or blocked, while linking each blocking concern to inspectable evidence and a responsible repair stage. We introduce a manifest-driven evaluation protocol with clean controls, counterfactual fault variants, base-bundle-disjoint splits, blind manuscript-only requests, and metrics for false promotion, concern–acceptance conflict, clean-case retention, repair localization, traceability, and cost. A synthetic development suite with four base bundles and 40 cases validates the evaluator and exposes three missing integrity checks in an artifact audit. After those checks were added, the audit passed all development cases; an advisory-only ablation still accepted every defective case despite detecting its concerns. These results are instrument validation, not empirical evidence for general effectiveness: the suite is synthetic, unreviewed, and explicitly ineligible for paper-level claims. We specify the independently adjudicated, real-provider experiment needed to test the central hypothesis.

1 Introduction

Research-agent systems span literature review, experimentation, and report writing (Schmidgall et al., 2025). Capability benchmarks now evaluate replication workflows extending from paper understanding to experiment execution (Starace et al., 2025). Completion, however, is not equivalent to scientific readiness. BadScientist shows that plausible but unsound papers can receive favorable automatic reviews (Jiang et al., 2026), while counterfactual evaluation finds that flaws in research logic have no significant effect on automatic reviews (Dycke and Gurevych, 2026).

We study the policy boundary between detecting a concern and allowing an artifact bundle to advance. A useful research governor must fail closed: a blocking concern cannot coexist with a paper-ready decision unless the concern is resolved or explicitly downgraded. It must also avoid the trivial solution of blocking everything. This motivates three questions: whether artifact-bound gates reduce false promotion, whether binding concerns to transitions prevents concern–acceptance conflict, and what clean-case, repair-localization, latency, and cost trade-offs follow.

The contribution at the present evidence level is a protocol and tested instrumentation, not a confirmed performance claim. We provide (i) a portable artifact-bundle task with explicit promotion and repair labels, (ii) leakage and adjudication gates, (iii) baselines that isolate detection from action binding, and (iv) a development failure analysis that routes audit misses back to the responsible workflow stages.

2 Related Work

Research-agent capability. Agent Laboratory organizes literature review, experimentation, and reporting within a multi-agent research workflow (Schmidgall et al., 2025). PaperBench evaluates replication of published machine-learning research from paper understanding through executed artifacts (Starace et al., 2025). These systems motivate artifact-level evaluation, but their primary object is research capability rather than the consistency of a promotion decision with detected blockers.

Review and claim verification. BadScientist operationalizes concern–acceptance conflict for unsound generated papers (Jiang et al., 2026). Dycke and Gurevych use targeted edits to create flawed research logic and compare automatic reviews (Dycke and Gurevych, 2026). CLAIM-BENCH evaluates claim extraction, evidence extraction, and claim–evidence link validation in AI papers (Javaji et al., 2025). These dimensions motivate a separate evaluation question: whether detected concerns constrain promotion decisions, bind artifact evidence, and identify the responsible repair stage.

Artifact assurance and recovery. ARA reconstructs scientific workflows for reproducibility assessment (Riehl et al., 2026). REPRO-Bench evaluates agents' reproducibility assessments using paper PDFs and reproduction packages (Hu et al., 2025). MADS-CPS proposes machine-checkable run-level admissibility for autonomous laboratories (Petel, 2026). SAGE routes experiment failures to structured intervention levels (Ma et al., 2026). ClaimGarden controls claim states under evidence updates (Nishi, 2026). The closest distinction is therefore not artifact inspection or failure attribution alone, but whether detected scientific blockers deterministically constrain paper-readiness promotion without over-blocking clean bundles.

3 Promotion Governance Protocol

3.1 Task and labels

Each case is a portable research-run artifact tree paired with a private case manifest. The evaluated system returns one decision from `promote`, `needs_review`, `downgrade`, or `block`; zero or more concerns with severity and evidence paths; and the workflow stages responsible for repair. Gold labels remain outside the visible artifact tree.

Every suite declares an evidence class, adjudication status, mutation-isolation status, and paper-claim eligibility. A paper-eligible suite must be independently double adjudicated and have double-verified mutation isolation. All variants derived from one base bundle remain in the same split, and source content hashes detect renamed duplicates across splits. The confirmatory intake freezer rejects fewer than 72 source-hash-distinct bases, duplicate source trees, insufficient source-family or operator-group diversity, and bundles that cannot support every registered mutation. It derives opaque base identities from source hashes and emits only held-out cases with provisional `needs_review` labels and `paper_claim_eligible=false`.

3.2 Fault families and controls

The protocol requires clean positive, null, and negative controls together with variants covering missing comparison evidence, repeated-run provenance, hidden failed execution, declared-versus-executed budget mismatch, result–figure conflict, claim–evidence conflict, citation support mismatch, stale persisted state, and unsupported claim strength. Each mutation records before and after hashes. The implementation exports each mutated artifact with its clean pair, declared family, and operations under an opaque audit ID. Two separate auditors must label every pair as isolated; either auditor can mark a pair confounded and block progression. The verifier binds records to suite, case, artifact, and mutation-manifest hashes. Promotion-label reviewers remain blind to mutation metadata, and their declared IDs cannot overlap the mutation auditor IDs.

3.3 Comparison conditions

We define four deterministic conditions and one provider condition. *Always promote* trusts the incoming ready state. A *presence checklist* verifies required files without semantic consistency checks. An *advisory artifact audit* reports the same concerns, evidence, and repair owners as the full audit but does not bind them to the decision. The *fail-closed artifact audit* applies the binding policy. Finally, a blind *manuscript-only reviewer* receives only paper text under opaque request identifiers; case labels, mutations, paths, and gold data remain private.

3.4 Metrics and inference

Primary metrics are false paper-ready promotion and concern–acceptance conflict. We also report four-way decision macro-F1, clean-case promotion, blocker precision/recall/F1, exact repair-owner match, trace coverage, latency, and provider cost. Every system trial must cover every case. Pairwise effects are computed per case, with 5,000-replicate bootstrap intervals clustered by base bundle and exact paired sign tests over base-bundle effects. Synthetic development suites remain exploratory regardless of numeric performance.

4 Implementation and Development Validation

The protocol is implemented as manifest-driven builders, runners, scorers, and blind request import/export commands. Paths are confined to suite roots; symbolic links, duplicate cases, hash mismatches, split leakage, and incomplete trial coverage are validation failures. Failure analysis emits

Condition	False	Conflict	Clean
Always promote	1.00	–	1.00
Presence checklist	1.00	–	1.00
Advisory audit	1.00	1.00	1.00
Fail-closed audit	.00	.00	1.00

Table 1: Synthetic development results over four base bundles and 40 cases. Columns report false-promotion, concern–acceptance conflict, and clean-case promotion rates. The labels were generated for instrument debugging and were not independently adjudicated; the table is not confirmatory evidence.

review diagnostics and node-strengthening recommendations with explicit recheck conditions.

The development corpus contains four synthetic base bundles, one clean control and nine counterfactual variants per bundle, for 40 cases. It is marked as synthetic development evidence with unreviewed adjudication and no paper-claim eligibility. Table 1 therefore tests instrumentation only.

Before strengthening, the fail-closed audit reached .70 decision accuracy and .333 false-promotion rate, missing repeated-run provenance, executed-budget, and stale-state faults. The failure analyzer routed the first two to experiment execution and the third to review. Added integrity checks were then evaluated on the same frozen suite. In the final development run, the fail-closed audit differed from the advisory ablation by .90 decision accuracy and -1.00 false promotion. The cluster bootstrap intervals are degenerate because each synthetic fault repeats identically across only four base bundles; the paired sign-test value is .125. This is direct evidence that the development sample is too small for inferential claims, despite final decision accuracy and blocker F1 of 1.00 on the generated cases.

5 Confirmatory Study and Readiness Gate

The frozen confirmatory study requires at least 72 source-hash-distinct base bundles and 720 held-out cases. Each base contributes one clean control and one variant from every registered fault family. The corpus must span at least three declared source families and three operator groups, with no group contributing more than half of the bases. Each base binds run configuration, events, metrics, review decision, command, and execution logs by hash. The intake gate rejects failed or non-real modes, fewer than three distinct source trials, hash drift, and duplicate run identities or execution fingerprints. This verifies the declared artifact record, not the real-world occurrence of execution. Two reviewers must adjudicate held-out promotion labels, and separate reviewers must verify mutation isolation without access to system predictions. Pseudonymous IDs enforce record-level role separation but do not prove real-world identity, so an external assignment log is also required. The manuscript-only condition requires three preserved real-provider trials per case. At least one post-repair rerun per fault family must measure recovery and regression. Raw requests, responses, decisions, failures, manifests, hashes, latency, and cost must be retained.

The central hypothesis is that fail-closed governance lowers false promotion by at least 20 absolute percentage points relative to the presence checklist while retaining at least 90 percent clean-case promotion. The study is stopped or downgraded if the full policy blocks more than 10 percent of clean bundles, if the checklist is within five points on both primary safety and clean-case metrics, if mutations leak labels, or if independent adjudication is absent.

6 Limitations and Ethical Considerations

Counterfactual faults may be easier than naturally occurring research defects, and deterministic checks can overfit a known artifact schema. A promotion gate does not establish scientific truth; it enforces consistency between available evidence, stated concerns, and allowed transitions. Provider-based reviewers introduce model, prompt, and temporal dependence. Public artifact bundles must exclude credentials, private paths, personal data, and proprietary research content. False promotion can amplify unsound claims, while false blocking can suppress valid null or negative results; both errors must remain visible.

7 Conclusion

Promotion governance treats paper readiness as an evidence-bound transition rather than a by-product of workflow completion. The implementation and development suite show that this distinction is measurable and that detected failures can drive targeted audit strengthening. They do not yet show that the policy generalizes. Independent artifact curation, adjudication, real-provider review, and a frozen confirmatory run are required before advancing beyond a research memo.

References

Nils Dycke and Iryna Gurevych. 2026. Automatic reviewers fail to detect faulty reasoning in research papers: A new counterfactual evaluation framework. *Transactions of the Association for Computational Linguistics*, 14:465–488.

Chuxuan Hu, Liyun Zhang, Yeji Lim, Aum Wadhwani, Austin Peters, and Daniel Kang. 2025. Repro-bench: Can agentic ai systems assess the reproducibility of social science research? *Preprint*, arXiv:2507.18901. ArXiv version v1, accessed 2026-07-16.

Shashidhar Reddy Javaji, Yupeng Cao, Haohang Li, Yangyang Yu, Nikhil Muralidhar, and Zining Zhu. 2025. Can AI validate science? benchmarking LLMs on claim →Evidence reasoning in AI papers. In *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*, pages 2355–2379, Mumbai, India. The Asian Federation of Natural Language Processing and The Association for Computational Linguistics.

Fengqing Jiang, Yichen Feng, Yuetai Li, Luyao Niu, Basel Alomair, and Radha Poovendran. 2026. BadScientist: Can a research agent write convincing but unsound papers that fool LLM reviewers? In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 24712–24727, San Diego, California, United States. Association for Computational Linguistics.

Jie Ma, Binfei Chu, Jie Gao, Jinlu Zhang, Yiwei Ma, Yi Tan, Jiayi Ji, Xiaoshuai Sun, and Rongrong Ji. 2026. One reflection is not enough: Self-correcting autonomous research via multi-hypothesis failure attribution. *Preprint*, arXiv:2606.31478. ArXiv version v1, accessed 2026-07-16.

Hafumi Nishi. 2026. ClaimGarden: Update-aware claim-state control for AI scientist workflows. In *ICML 2026 Workshop on AI for Science*.

Mateo Petel. 2026. MADS-CPS: A machine-checkable admissibility contract for AI scientists in autonomous laboratories. In *ICML 2026 Workshop on AI for Science*.

Kevin Riehl, Andres L. Marin, Nikofors Zacharof, Fan Wu, Patrick Langer, Robert Jakob, Anastasios Kouvelas, Georgios Fontaras, and Michail A. Makridis. 2026. Ara: Agentic reproducibility assessment for scalable support of scientific peer-review. *Preprint*, arXiv:2605.02651. ArXiv version v2, accessed 2026-07-16.

Samuel Schmidgall, Yusheng Su, Ze Wang, Ximeng Sun, Jialian Wu, Xiaodong Yu, Jiang Liu, Michael Moor, Zicheng Liu, and Emad Barsoum. 2025. Agent laboratory: Using LLM agents as research assistants. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 5977–6043, Suzhou, China. Association for Computational Linguistics.

Giulio Starace, Oliver Jaffe, Dane Sherburn, James Aung, Jun Shern Chan, Leon Maksin, Rachel Dias, Evan Mays, Benjamin Kinsella, Wyatt Thompson, Johannes Heidecke, Amelia Glaese, and Tejal Patwardhan. 2025. Paperbench: Evaluating ai's ability to replicate ai research. *Preprint*, arXiv:2504.01848. ArXiv version v3, accessed 2026-07-16.